A thousand tiny hooks

How Velcro Works

Name _____ Date _____



Little Wonderer pulls open a shoe strap. Rrrip! The little one presses it shut, and it holds tight again. Big Wonderer leans in. The strap has two sides, one rough and one soft. People call it Velcro, but that is a company name. Its plain name is hook and loop. What is hiding in that sound?



In 1941, a Swiss engineer went for a walk with his dog. Afterward, burdock burrs clung to his socks and the dog's fur. He looked at a burr under a microscope. Each spine ended in a tiny hook. The hooks caught on loops of thread and fur. Could cloth copy that trick?



Look very closely at the two sides. The rough side holds rows of stiff little hooks, like tiny bent canes. The soft side is a fuzzy tangle of thread loops. Press them together, and hundreds of hooks catch hundreds of loops. One hook is weak, but many together are strong. The name Velcro mixes French words for velvet and hook.

hook and loop	a strap with tiny hooks on one side and soft loops on the other
burr	a prickly seed case that sticks to fur and socks
microscope	a tool that makes tiny things look big
shear	a sideways, sliding pull
lint	bits of fluff from cloth

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Try pulling the two sides apart sideways, like sliding. It is very hard, because every hook pulls at the same time. Now peel back one corner. It opens easily, because only one thin line of hooks lets go at a time. The rrrip is hooks springing free, row after row. A sideways pull is called shear.



A strip on a little board holds a small cloth bag. Big Wonderer points at it. The bag pulls straight down along the strip, so all the hooks share the load, and nothing slips. Then Little Wonderer reaches up and peels one corner. Rrrip! A few hooks at a time let go, and the bag slides off.



Making the hooks took years of trying. The inventor wove nylon loops, heated them to set their shape, then snipped one side of each loop to leave a hook. After many uses, lint fills the hooks, so they cannot reach the loops. A fine comb can lift the fuzz out. Why would a cleaner strip grip better?

1. What did the engineer see on the burr?
2. What is on the rough side, and what is on the soft side?
3. Why is a sideways pull harder than a peel?
4. Why does a strip full of lint grip less?

Your turn

Write or dictate three sentences: where the idea came from, what the hooks catch, and why a peel opens easily.

Example: The idea came from a burr. The hooks catch soft loops. A peel opens a few hooks at a time.